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Calcium Phosphate Powder for Cancer Vaccination

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Abstract. Cancer cells synthesize abnormal proteins and peptides which are associated to heat shock proteins being overproduced by these cells due to the stress induced by the particular biology of cancer tissue. We have purified on hydroxylapatite powder heat shock proteins using the HA-particles as purification bed, vectors for the proteins and vaccination adjuvant. The powder make possible that the purified HSPs and their associated peptides are transfected to the antigen presenting cells and presented to the T cells for the destruction of the cancer cells bearing the antigens.

Introduction

Human subjects are able to mount a T cell mediated response to cancer based on their recognition of epitopes from mutated proteins, the products of overexpressed oncoproteins or ectopically expressed developmental genes.

For various reasons, the abnormal proteins synthesized by the cancerous cells are not recognized by the immune system as they should be. The aim of active immunotherapy is to make these proteins recognizable by the T cells.

When recognized, the tumor antigens are internalized by the antigen presenting cells (APCs) and then presented at the cell surface to the T cells (CD8+) which will destroy the cancer cells carrying the antigens. They are, at first, endocytosed into APCs within endosomes which fuse with Golgi vesicles in which the acid environment partially degrades the endocytosed antigens. The late endosomes containing these degraded elements can then fuse with the vesicles in which the MHC-I or II peptides are transferred at the cell surface. Inside these compartments, the peptides issued by the tumor cell antigens having an affinity for the MHC-I/II peptides constitute a complex which will be positioned at the cell surface.

The antigen presenting cells originate from the monocyte cells. They are professional phagocytes and can be macrophages or dendritic cells.

The T cells (CD8+) can recognize and kill the tumor cells when tumor cell antigens are processed and expressed at the level of MHC-I peptide. In order that these cells are efficient against the tumor cells, when they enter in contact with the target cells, there must be complementary adherence molecule exposed at the surface of the two cells. The T cell response takes place in the lymph node. The antigen presented at the MHC II level will be recognized by the CD4+ T lymphocytes. They are activated only if a co-stimulating signal is present and which is transmitted by the B7 molecules. The amount of co-stimulating molecules is correlated with the maturation state of the APCs.

The heat shock proteins (HSPs) are a class of proteins synthesized when the cells are submitted to a stress and in particular to heat. They stabilize other cell protein folding and thus are intimately associated to these late proteins. They also have other functions. In particular, some of them such as gp96 or HSP 70 are involved in the process of T-cell cross-priming by antigen presenting cells.

As the HSPs are associated to the peptides synthesized by the cell, the purification of these proteins produced by the cancer cells allows to get a good finger print of the cancer cell peptide synthesis.

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The cancer cells synthesize a high amount of HSPs as they are stressed by the abnormal tissue biology. Due to their associated peptides, they are patient and tumor specific. They also depend on the time of purification during the tumor development due to the genetic instability of the cancer cells.
HSPs complex can interact specifically with receptors located at the APCs surface and are taken up by these cells. The peptides carried on HSPs are then processed and delivered to the cell surface in association with MHC class I antigens for recognition by antigen-specific $CD8^+$ T cells. Thus tumor HSPs can be used to make some of the associated peptides recognised by the T cells. The aim of this experimentation was to develop a fast way for HSP purification using HA-ceramic powder and to load HA- particles with autologous HSPs and their associated proteins in order to reinject the particles in the patient to stimulate its immune system against own patient tumor.

Materials and Methods

We have performed antigen purification from biopsies obtained during surgical operation for tumor removal (gliomas, kidney adenocarcinomas, breast cancer, parotidis cancer).

Table 1: characteristics of the powder:

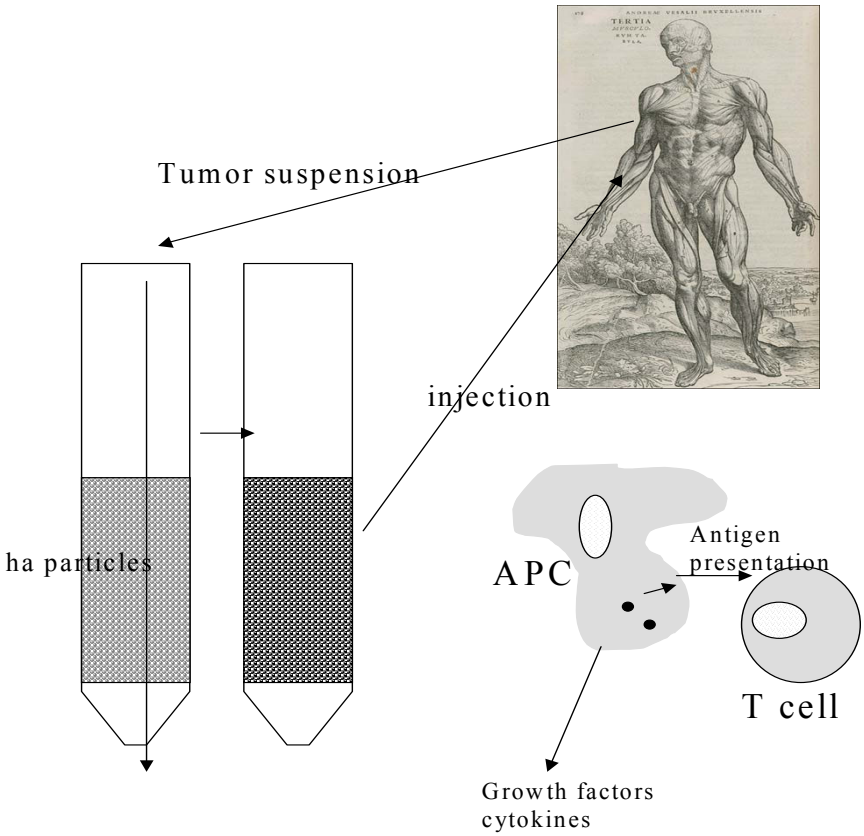
Powder	P15
Nature of the charged groups	PO_4^{3-}, OH^-, Ca^{2+}
Electrocinetical potential (mV)	-35
Hydrophobicity	+
Surface pH	7,8
Granulometry (µm)	80-125
Surface area (m^2/gr)	0,62
Shape	spherical

The purification of the proteins on an HA-powder is performed from the soluble part of the grounded tumor. The solution can be percolated through the column at atmospheric or high pressure. When the process is carried out under atmosphere, the granulometry of the powder is an important factor in order to avoid any plugging of the column.
The particles must be washed by a 200 mM PBS at ph 7 in order to reduce the contamination by un wanted proteins.
At this stage, a powder suspension is performed and co-factors can be added to the suspension such as cytokines, interferon or interleukines. We use as co-factors the tumor cell membranes obtained by centrifugation. The powder suspension is then injected subcutaneously in the patient (one injection weekly during one month, 150 µg of protein at every injection).
We have realized electrophoresis of the fractions collected after washing the column with various NaCl concentration (fig.2).

Results and Discussion

This technique allows to obtain adsorbed at the particle surface: HSPs (gp 96 and HSP 70 and 27), β-catenin, Her-2/neu, P-cadherin, and surviving. Several of these proteins have shown that they can be used for vaccination against several types of tumors in human beings [1]. No toxicity or secondary effects were reported when used against melanoma, colorectal and kidney adenocarcinoma, breast cancers or gliomas [2]. More interestingly, these proteins trigger an activation of the CD8+ cells against the cancer cells [3].

Figure 1: the tumor suspension from which the cell debris were removed is percolated through the HA-column. The loaded particles are then subcutaneously injected in the patient in which they trigger the APCs.



Ceramic particles adsorb less proteins per mass unit than encapsulating particles such as PLA. However, the amount of antigen necessary to stimulate the immune system is low and compatible with the amount carried by the HA-particles.

We have demonstrated that the same particles, when loaded with plasmids carrying a LacZ gene, can transfect preferentially the cells of the monocyte lineage [4]. The phagocytosis of the particles by these cells also lead to their activation making them to synthesize different cytokines and growth factors stimulating the immune system [5].

The surface properties of the particles are particularly interesting because they permit adsorption of these proteins and their release at a ionic strength and a pH which is close to that encountered in the intracellular medium.

The injection of the particle suspension in the subcutaneous tissue trigger an infiltration by monocytes and macrophages which is of particular interest for the activation of APCs [2]. This particularity combined to other such as protein adsorption, ceramic degradation, particle phagocytosis, can be exploited both to vectorize the proteins into APCs and stimulate them.

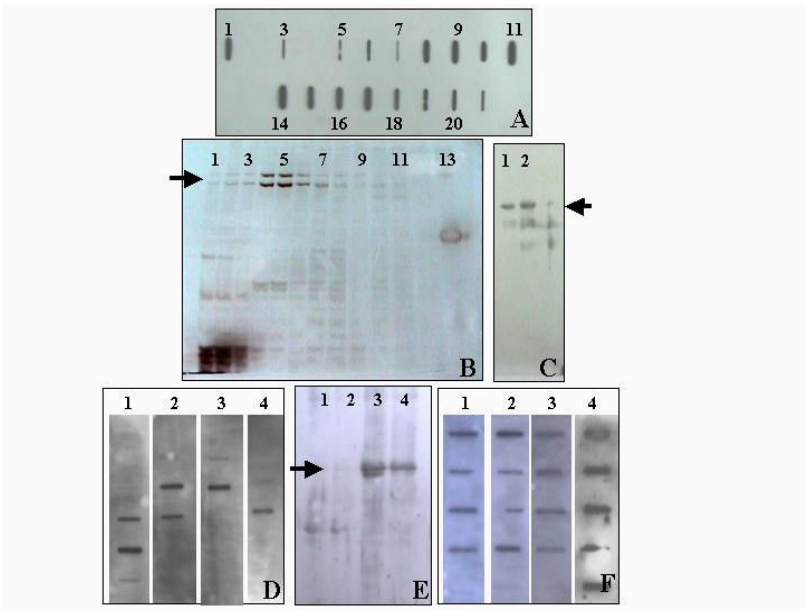


Figure 2: **Purification of the vaccine components.** A: **slot blot to reveal gp96 after HA column chromatography. Lane 1: positive control. Lane 2: negative control. Lane3: flow through after column equilibration. Lanes 4-6: fractions collected after column washing with 50 mM NaCl. Lanes 7-9: fractions collected after column washing with 100 mM NaCl. Lanes 10, 11 and 14: fractions collected after column washing with 200 mM NaCl. Lanes 15-17: fractions collected after column washing with 300 mM NaCl. Lanes 18-20: fractions collected after column washing with 400 mM NaCl. Lanes 21, 22: fractions collected after column washing with 500 mM NaCl.** B: **SDS-PAGE and silver staining of the fractions collected from the HA chromatography column. Note that the fractions collected with 200 mM of NaCl contained the highest amount of gp96 (arrow) which appeared as a doublet (lanes 4-6). The other lanes were loaded with fractions obtained washing with lower and higher mM of NaCl concentrations (as shown in the slot blot).** C: **western blot showing gp96 as a doublet (arrow) after HA purification.** D: **slot blots of the gp96 purification fractions to show examples of the identified proteins. 1: β-catenin, 2: Her-2/neu, 3: P-cadherin, 4: survivin.** E: **western blot to reveal Hsp70 after membrane purification with sucrose gradient centrifugation. Lanes 1 and 2: loaded with 20 and 10 μg of proteins obtained from a patient with a parotid adenocarcinoma. Lanes 3 and 4: loaded with 20 and 10 μg of proteins obtained from a patient with a renal clear cell carcinoma. Note that lane 3, loaded with a higher protein concentration, shows two Hsp70 bands: the upper corresponds to the constitutive form and the lower corresponds to the inducible form of Hsp70.** F: **slot blots of the purified membranes to show examples of the identified proteins. 1: Her-2/neu, 2: Hsp27, 3: P-cadherin, 4: β-catenin.**

Conclusions

The same ceramic particles can be used for purification, vectorisation, transfection and as adjuvant in the cancer vaccination field. These particles must have special characteristics also allowing the percolation of the tumor suspension trough the powder and the powder suspension in liquid in order to be injected. The first clinical trials demonstrated that this vaccination was safe and did not show systemic reaction. This technology can be applied to a lot of tumor types and specially to tumors for which no specific proteins were identified.

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